RoboCloth: A Large-Scale Real Cloth Material Dataset for Neural Reflectance Reconstruction

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https://theialab.github.io/robocloth

Figure 1. We design a robotic system to capture a large dataset of fabric materials, so to learn a neural BRDF representation. *Left*: our capture system, two UFACTORY X-ARM 6 arms carrying a camera and a light source, with an example captured image shown in the inset. The photo of the rig is over-exposed so to be able to see the robotic arms, so the light saturates; captures use short exposures within the sensor's linear range. *Right*: materials fitted via our learned prior on the test-set, yielding neural BRDFs that can be photo-realistically rendered with path-tracing (i.e. the BRDFs of the fabric on the sofa, curtains, pillows, and carpet).

Abstract

We present RoboCloth, a dataset of 500 real cloth materials captured under flexible, calibrated camera and lighting configurations, where both are robotically controlled during acquisition. Each material is imaged in approximately 580 high-dynamic range frames spanning a wide view-light hemisphere; camera poses are refined via structure-from-motion and aligned to the robot frame, yielding per-material sparse surface geometry and structured per-point reflectance observations ready for neural material training. We demonstrate the value of this data through a two-stage neural reconstruction pipeline: Stage 1 trains a shared MLP decoder on our corpus to learn cloth material priors; Stage 2 then freezes the decoder and fits a dense 2048×2048 *latent texture to each target material. Evaluating on held-out materials from our own dataset and on the independent UBO2014 benchmark, we show that a decoder trained on our captures outperforms decoders trained on existing measured reflectance datasets and analytic PBR fitting, transferring to unseen acquisition systems without modification. The dataset, calibration files, observation tensors, and train/test splits are released under CC BY 4.0 on the Hugging Face Hub.*

1. Introduction

Cloth is challenging to render faithfully, as real fabrics exhibit spatially varying reflectance driven by fiber microstructure, weave geometry, surface roughness, and anisotropic scattering — properties that interact nonlinearly with viewing and illumination direction and vary continuously across the surface. Recovering dense, shader-ready material representations from real observations is a prerequisite for photo-realistic cloth rendering in physically based pipelines, with direct applications in virtual try-on, digital fashion, and visual effects. Recent work has shown that learning a *shared* neural BRDF decoder across many materials enables generalizable reflectance estimation without committing to any analytical BRDF parameterization [24, 27]. In this paradigm, a decoder maps compact per-point latent codes to reflectance values queryable by a path tracer [23, Ch. 14]; given a new

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material, a dense latent grid is optimized against sparse observations with the decoder frozen, producing spatially varying reflectance maps that plug directly into a physically based rendering pipeline. The decoder amortizes reflectance estimation across materials: a prior trained on sufficiently diverse and well-covered captures can generalize to unseen fabrics and recover dense representations from limited measurements, a capability that neither single material overfitting nor purely analytical models can offer.

The limiting factor is training data. Learning a generalizing decoder for cloth requires captures that simultaneously satisfy three conditions: *real* measured appearance to avoid real-to-synth domain gaps, *diversity* in the captured materials to support generalization, and *dense+flexible* view-light coverage—directions sampled freely across both hemispheres rather than from a preset fixed hardware layout—to enable reliable reflectance-geometry disentanglement. Existing resources do not combine real cloth diversity with flexible view-light coverage at scale; see Table 1. Most large-scale datasets of real materials are not multi-view: each surface is observed from only one or two camera positions [17, 46]. Multi-view alternatives are either not spatially varying [5, 18], or cover fewer than a hundred materials [39]. The closest resource to ours, UBOFAB19 [20], provides 378 real fabrics with multi-view, spatially varying measurements, but its view and light directions are dictated by a preset fixed hardware layout rather than sampled freely across both hemispheres, which bounds the angular coverage of the view-light domain. The cross-dataset results in Section 5.3 show the performance drop caused by this insufficient view-light coverage.

To fulfill these requirements, we present RoboCloth, a system designed to capture a dataset of 500 materials. Our system comprises two robot arms carrying an HDR camera and a calibrated LED area light, together with a motorized turntable, yielding ≈ 580 HDR images per material; see Figure 1. View and light directions can be sampled freely across both hemispheres, rather than being constrained to a fixed set of hardware-defined configurations. While approximate light and camera positions are controlled by the robot API, we recover accurate poses and sparse surface geometry via COLMAP [30].

Our dataset is not merely a larger image collection; it directly targets a downstream task: recovering dense, shader-ready material representations from real cloth captures for a path-traced pipeline. To this end, we train a neural BRDF decoder on our data and evaluate it by fitting a dense per-pixel material map (a latent texture) to each held-out material, measuring reconstruction accuracy and appearance under path-traced rendering, and comparing against classical PBR fitting [2] and decoders trained on existing material datasets. A controlled ablation over dataset scale quantifies its contribution to downstream generalization.

Dataset	Publication Year	Dataset Scale	Cloth Specific	Real World	Spatially Varying	Multiple Views	Flexible Coverage
MERL [18]	2003	100	✗	✓	✗	✓	✗
UTIA [5]	2014	150	✗	✓	✗	✓	✗
UBO2014 [39]	2014	84	✗	✓	✓	✓	✗
UBOFAB19 [20]	2019	378	✓	✓	✓	✓	✗
Real-SVBRDF [1]	2020	80	✗	✓	✓	✗	✗
PhotoMat [46]	2023	12,000	✗	✓	✓	✗	✗
UMat [28]	2023	2,000	✓	✓	✓	✗	✗
OpenSVBRDF [17]	2023	1,000	✗	✓	✓	✗	✗
MatSynth [36]	2024	4,069	✗	✗	✓	✗	✗
RoboCloth	–	**500**	✓	✓	✓	✓	✓

Table 1. Comparison wrt. related works. We report whether each dataset is cloth-specific and provides real measurements, spatially varying appearance, multiple views, and flexible view-light coverage. With multiple views, we denote capture with more than two distinct viewing directions. RoboCloth is the only dataset to provide all five, and at scale.

To summarize, our contributions are: (1) A real cloth reflectance dataset: 500 fabric materials with $\approx$580 HDR images each, under flexible view-light coverage, with preprocessed data, splits, and loaders for neural BRDF training; see Section 3. (2) A calibrated robotic acquisition pipeline: an end-to-end capture and processing system producing aligned HDR observations, camera and light metadata, COLMAP-refined poses, and sparse surface reconstructions; see Section 3 and Section 9. (3) A neural decoder method for cloth material reconstruction: training and evaluation of a neural BRDF decoder by fitting a dense per-pixel latent texture to each held-out material, with path-traced rendering evaluation and an ablation over dataset scale; see Section 5.

2. Related Work

We group prior work into four categories: measured reflectance and material datasets, inverse-rendering and relighting datasets, neural material representations, and calibrated capture systems.

Measured reflectance and material datasets. Classical BRDF databases such as MERL [18] (100 isotropic materials) provide dense per-sample reflectance but no spatial variation across the surface. Denser angular collections are either not spatially varying or fixed to specialized hardware: the UTIA anisotropic BRDF database [5] (150 materials, 96 fabrics) stores one spatially-averaged BRDF per material rather than a per-point SVBRDF. Although its acquisition setup uses actuated camera and light arms, their motion is constrained to fixed-radius goniometric trajectories rather than arbitrary Cartesian poses, and all materials share the same uniform 81×81 directional grid. The UBO2014 BTF database [39] (84 materials, 22,801 images each) captures per-point appearance only under a fixed dome. Among cloth-specific resources, UBOFAB19 [20] provides 378 fabric materials acquired with a commercial X-Rite

TAC7 scanner under a preset fixed hardware layout, providing at most 768 calibrated HDR measurements per material and Ward anisotropic SVBRDF fits in AxF format. Tang et al. [31] capture woven-fabric BSDFs from a reflection-transmission photo pair. UMat [28] provides 2,000 real textiles for single-image high-resolution material capture, while OpenSVBRDF [17] scales to 1,000 near-planar SVBRDFs using a 16,384-LED rig and only two camera views ($\approx$193 raw HDR images, $\approx$15 min per sample), but is not cloth-specific. Real-SVBRDF [1] covers 80 real materials across 462 capture sets with only 12 images per set. Across these measured resources, none targets a large cloth corpus with flexible view-light coverage (see Kavoosighafi et al. [10] for a broader survey). Moreover, none of these works compares against other datasets by evaluating how well a prior trained on its data generalizes to new materials. On the synthetic side, MatSynth [36] (4,069 PBR materials) offers strong pre-training but does not replace real, calibrated cloth captures.

Inverse rendering and relighting datasets. Several datasets provide real object-level multi-illumination benchmarks useful for evaluating inverse rendering and relighting [12, 15, 22, 35, 47]. But these target *object-centric* capture, and are neither designed for learning cloth material priors, nor for evaluating dense shader-ready material reconstruction from near-planar samples.

Neural material representations. NeuMIP [13] represents materials as multiscale neural feature textures and has become a standard backbone for learned material models, with subsequent work extending neural reflectance representations through hierarchical architectures, efficient neural sampling, real-time appearance models, and spherically distributed primitives [4, 40, 42–44]. Unified neural BTF encoding [27], building on neural BTF compression [25], trains a decoder shared across materials and supports encoding previously unseen materials. HyperBRDF [7] and differentiable rendering pipelines [3, 14] show that reconstruction quality is tightly coupled to training data diversity and angular coverage. PhotoMat [46] trains a strong generator from 12,000 single-flash real photos, underscoring the value of real captured data. Generative Neural Materials [24], diffusion-based generators, and single-image capture methods [28, 37, 38, 45] all depend on large, diverse, and realistic training corpora. Despite this progress, real captured datasets with cloth-specific diversity and flexible view-light coverage remain limited.

Calibrated material capture systems. Capture systems trade off hardware, coverage, and throughput: dense domes [39] and LED arrays [17] give rich angular sampling but need dedicated, non-portable hardware. Robot-based gonio-reflectometers such as ARGon3 [8] automate dense angular measurements of individually mounted samples. Closest to our design is the BTF scanner of Rainer et al. [26], which moves a camera and an LED light on two arms around a static sample and acquires a dense 40×40 view-light scan in about 11 hours per sample; its measurements were not publicly released. Commercial scanners [20] provide high-quality fabric measurements under a preset fixed hardware layout. Lightweight rigs [1] reduce hardware cost but yield sparse per-sample observations. Our robotic system—two robot arms carrying one HDR camera and one calibrated LED area light, together with a motorized turntable—enables practical cloth acquisition with hundreds of calibrated images per sample under flexible view-light variation, without commitment to a fixed hardware geometry or analytical BRDF model.

3. Dataset

RoboCloth is a dataset of 500 real cloth materials, each captured under approximately 580 calibrated camera-light configurations using a robotic acquisition rig. For each material we release 16-bit linear HDR images, robot-frame-aligned sparse geometry, a structured per-point observation tensor, and the calibration metadata required to reproduce our rendering pipeline. The dataset is split into training ($\approx$400 materials) and test ($\approx$100) partitions and is hosted on the Hugging Face Hub under `CC BY 4.0`. Full construction details and ablations are provided in Section 9.

3.1. Acquisition

As illustrated in Figure 1, our robotic rig consists of two UFACTORY XARM 6 robot arms (one carrying the camera, one carrying an LED light source) and a ferromagnetic turntable inside a matte-black enclosure. The turntable provides a third controlled degree of freedom: rotating the sample re-orients it beneath the fixed camera-light pair, so each arm configuration is reused across several distinct sample-relative azimuths, effectively tripling the diversity of camera-light configurations reachable without moving the arms. Each cloth sample is fixed flat by corner magnets (standardizing the flattening procedure across operators), and a ChArUco fiducial [6, 29] mounted beside the sample serves for both offline calibration and per-session drift checks. The sample width and length are recorded at mounting time and later used to crop the valid material region from the sparse point cloud. Each material is acquired in a single automated session lasting approximately 20 minutes. At each of the $\approx$580 steps, the system records one raw Bayer HDR frame together with the robot-logged camera pose, light pose, and turntable angle, giving a densely paired image-and-metadata stream with no manual annotation. Each pixel then contributes a per-point radiance observation: In Section 3.2 we describe how to place surface points, cameras, and lights in a common metric world frame, and in Section 3.3 we calibrate the observed radiance to physical units.

3.2. Geometry calibration and reconstruction

To place every surface point, camera, and light in a single metric frame, a one-time hand-eye calibration [32] recovers the camera-to-flange transform ${}^{g}T_{c}$, and a separate procedure fits the turntable rotation axis in the robot base frame from checkerboard-target images. Composing these with the robot-logged arm pose gives each frame's camera pose in a common metric $0°$-turntable world frame:

$$ {}^{w_0}T_{c,i} = {}^{w_0}T_{b}(\theta_i)\, {}^{b}T_{g,i}\, {}^{g}T_{c}, $$

where ${}^{w_0}T_{b}(\theta_i)$ is the Rodrigues rotation that undoes the turntable angle θ_i about the calibrated axis, and the arm pose ${}^{b}T_{g,i}$ is read directly from the robot controller's forward kinematics (its API) rather than estimated. To recover a sparse point cloud, we employ COLMAP [30] on low-dynamic-range (LDR) copies of the frames. Because the COLMAP reconstruction is defined only up to a similarity, we align it to the metric world frame in closed form via Umeyama [34] using matched camera centers as correspondences. Frames whose alignment residual exceeds $16\,\mathrm{mm}$ are discarded, and the sparse cloud is cropped to the known sample footprint and filtered in depth by the Tukey IQR rule [33].

3.3. Radiometric processing

To make the observed radiance physically meaningful, raw Bayer mosaics are de-mosaiced into linear-light HDR RGB using Menon et al. [19]; no gamma or tone mapping is applied, preserving the raw radiance values recorded by the sensor. We additionally calibrate a per-exposure camera factor κ (a single scalar that converts pixel values to physical radiance under our linear sensor model) and the LED directional profile $\Lambda(\theta)$[1]. A 3×3 color correction matrix is estimated from a ColorChecker capture and reported alongside the release; the distributed images ship as white-balanced sensor RGB without the matrix applied, so downstream users retain full colorimetric control.

3.4. Observation tensor

Our training representation bypasses per-frame image loading entirely. We project each retained sparse point into every retained HDR frame and bilinearly sample the corresponding RGB value, assembling a dense tensor of *surface points* $\mathbf{O}_m \in \mathbb{R}^{K_m \times V_m \times 3}$, where K_m is the number of registered frames and V_m the number of points. Zero entries encode missing observations; no separate visibility mask is needed. After filtering, each material yields on average $\approx 1.6\times10^{8}$ valid RGB observations—a compact, memory-resident representation that replaces the ≈6–7 GB of per-material HDR frames (and their repeated per-frame decoding) with a single $\approx 1.0\,\mathrm{GB}$ indexed tensor of measurements.

[1] A 90-entry polar table measured from a near-Lambertian reference card, which will be later used as L_e^{Λ} in (1)

4. Neural Reconstruction Pipeline

We evaluate our dataset through a two-stage neural reconstruction pipeline, as illustrated in Figures 2 and 3. Stage 1 trains a shared MLP decoder on the full training corpus to learn a material prior. Stage 2 freezes that decoder and fits a dense per-material latent texture to each unseen cloth. This freeze-and-fit scheme is similar to Generative Neural Materials [24], which trains a shared decoder on synthetic materials and fits per-material latent textures against the frozen decoder to build training data for generative modeling. The decoder architecture, renderer, and the Stage 2 training step (losses, optimizer, and training schedule) are held fixed across all comparisons.

4.1. Differentiable Renderer

Both stages share the same differentiable single-bounce [9] path tracer, adopting the PyTorch differentiable path-tracer implementation from FIPT [41]. Indirect illumination is omitted, as the sample sits in a matte-black enclosure with no secondary radiant sources within the camera's dynamic range. For a surface point $\mathbf{x}$ with outward normal $\mathbf{n}$, the outgoing radiance toward the camera is

$$ L_o(\mathbf{x}, \boldsymbol{\omega}_o) = \int_{\mathcal{E}} f_r(\mathbf{x}, \boldsymbol{\omega}_i, \boldsymbol{\omega}_o) \cdot L_e^{\Lambda}(\mathbf{y}, -\boldsymbol{\omega}_i) \cdot V(\mathbf{x}, \mathbf{y}) \, \frac{(\boldsymbol{\omega}_i \cdot \mathbf{n})}{\|\mathbf{y} - \mathbf{x}\|^2} \, \mathrm{d}A(\mathbf{y}), \quad (1) $$

where $L_e^{\Lambda}(\mathbf{y}, -\boldsymbol{\omega}_i) = L_e\, \Lambda(\theta)$ is the constant LED radiance shaped by the calibrated directional falloff $\Lambda(\theta)$ (Section 3.3), and the remaining terms account for geometry and visibility. Because $\Lambda(\theta)$ already encodes the emitter's angular falloff, no separate light-side cosine appears in the integrand. The finite-area integral is approximated with four Monte Carlo samples per surface point ($S=4$), which suffice given the LED's small size relative to the working distance. The key design choice is that the neural decoder predicts *only* f_r, while cosine and geometry factors, emitter radiance, and inverse-square fall-off are all computed analytically from calibrated quantities. This separation keeps the learned representation physically grounded and makes the frozen decoder directly portable to standard physically based renderers without any modifications.

4.2. Stage 1: Learning a Material Prior

Stage 1 trains a single MLP decoder D_θ that, conditioned on a per-point latent code and local directions, predicts the BRDF value at any queried configuration. We adopt an auto-decoder formulation: each sparse surface point j of each training material m owns three directly optimizable quantities stored in aligned per-point parameter tables—a latent code $\mathbf{z}_{m,j} \in \mathbb{R}^{24}$, a surface normal $\mathbf{n}_{m,j} \in \mathbb{R}^3$, and a tangent vector $\mathbf{t}_{m,j} \in \mathbb{R}^3$ — initialized to $(0, 0, 1)$ and $(0, 1, 0)$

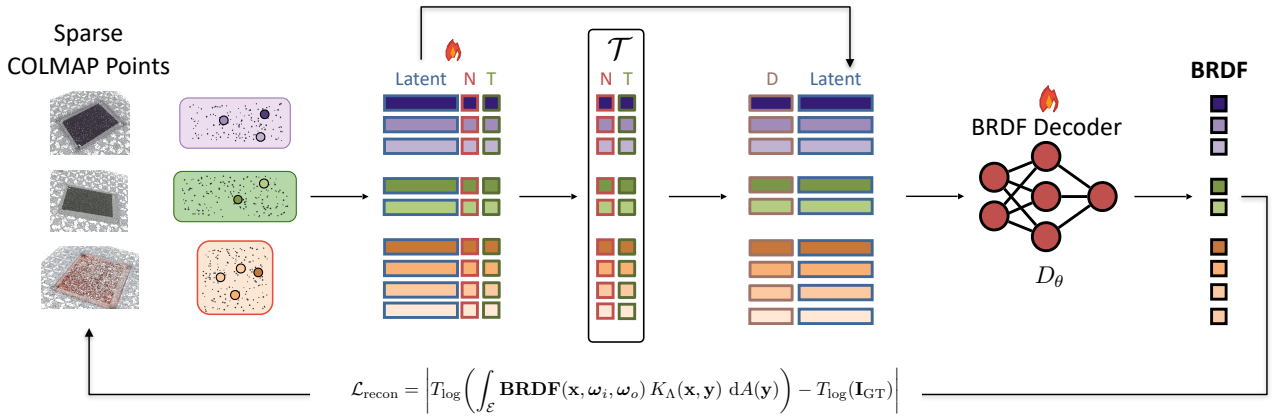

Figure 2. **Stage 1 — learning a material prior.** A single BRDF decoder, shared across all training materials, is optimized jointly (flame) with per-point tables of latent codes, normals, and tangents; see Section 4.2. N: normal, T: tangent, D: encoded local directions; each color denotes one material, with a few points drawn per material in every minibatch.

respectively, consistent with the flat mounting geometry. Before the decoder, a shared *coordinate-transform operator* $\mathcal{T}$ (Figure 2) maps each point's geometry and directions into the decoder's input: it forms a per-point tangent-bitangent-normal (TBN) frame from $(\mathbf{n}, \mathbf{t})$ by Gram–Schmidt, rotates the incident and outgoing directions $\boldsymbol{\omega}_i, \boldsymbol{\omega}_o$ into that frame, encodes the local directions and normal with degree-3 real spherical harmonics, and concatenates those encodings with the latent code $\mathbf{z}$. The same $\mathcal{T}$ is used in both stages; its exact construction is given in Section 10 and Figure 6. The decoder D_θ is shared across all materials and all points; the per-point parameter tables $(\mathbf{z}, \mathbf{n}, \mathbf{t})$ specialize it to each surface location and are updated by the same optimizer step as θ. Training in this stage is *sparse* and *multi-material*: each iteration draws a batch of (material, point, view) triplets sparsely sampled across many different training materials, so the single shared decoder is exposed to the full diversity of the corpus. We evaluate (1) at each sampled observation using the corresponding TBN frame, and back-propagate into both θ and the per-point parameter-table entries. We learn our priors by optimizing the training objective: $\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_{\text{decay}}\, \mathcal{L}_{\text{decay}}$, whose terms we will now detail.

Reconstruction loss. As HDR captures span four to five orders of magnitude in radiance, and a standard ℓ_1 loss would be dominated by specular highlights, we use a log-relative reconstruction loss, following the log-domain mapping introduced for BRDF learning [21]:

$$\mathcal{L}_{\text{recon}} = \left| \log\left(\frac{\hat{\mathbf{I}} + \varepsilon}{\rho_{\text{ref}} + \varepsilon} + 1 \right) - \log\left(\frac{\mathbf{I}_{\text{GT}} + \varepsilon}{\rho_{\text{ref}} + \varepsilon} + 1 \right) \right|, \quad (2)$$

where $\hat{\mathbf{I}}$ is the rendered image obtained by evaluating (1) at the observed configurations, $\mathbf{I}_{\text{GT}}$ is the captured HDR image, ρ_{ref} a per-dataset reference scale (Section 12), and ε a small constant for numerical stability; we write this *log-relative* transform as $T_{\log}(\cdot)$. This gives equal weight to dark diffuse regions and bright specular highlights across the full dynamic range.

Decay loss. Since captured observations do not cover incidence angles near grazing ($\cos\theta_i \approx 0$), the decoder is unconstrained in that region. Physically, the contribution $f_r \cos\theta_i$ must vanish as $\theta_i \to 90°$. At each iteration, for a shared azimuth, we sample two incident directions $\boldsymbol{\omega}_i^h$ and $\boldsymbol{\omega}_i^\ell$ with incidence-angle cosines c_h and $c_\ell = \alpha\, c_h$ ($\alpha \in [0, 1]$): the anchor c_h is drawn from $[c_{\min}, c_{\max}]$, just inside the most grazing observed angle, and $c_\ell \le c_h$ places the second sample further toward grazing (90°), in the under-constrained region. Defining $q(\boldsymbol{\omega}_i) = f_r(\boldsymbol{\omega}_i, \boldsymbol{\omega}_o) \cos\theta_i$, we penalize $q(\boldsymbol{\omega}_i^\ell)$ whenever it exceeds the linear interpolation between a value of zero at grazing, and the anchor $q(\boldsymbol{\omega}_i^h)$:

$$\mathcal{L}_{\text{decay}} = \Big[T_{\log}(q^\ell) \;-\; T_{\log}\big(\alpha\, \overline{q^h}\big) \Big]_+, \quad (3)$$

where $T_{\log}$ is the same log-relative transform as in (2), $\overline{(\cdot)}$ denotes stop-gradient, and $[\cdot]_+$ is ReLU. We ablate this loss against alternatives in Section 7. The stop-gradient treats the anchor contribution q^h as a fixed target, so the loss only suppresses the more-grazing prediction q^ℓ. This encourages $f_r \cos\theta_i$ to decay as a concave function towards zero as we approach grazing angles, yet without forcing the raw BRDF itself to vanish.

4.3. Stage 2: Dense Material Reconstruction

While Stage 1 operates on sparse triplets $(\mathbf{z}, \mathbf{n}, \mathbf{t})$ provided by COLMAP points of multiple materials, Stage 2 is *dense* and operates on a *single* material. For one target cloth, in Stage 2 we freeze D_θ and densely optimize a 2048×2048 latent-geometry grid $\mathbf{Z}$ covering the full surface, where each texel stores the triplets alike those in Stage 1—latent code, normal, a tangent. Every retrieved triplet is passed through

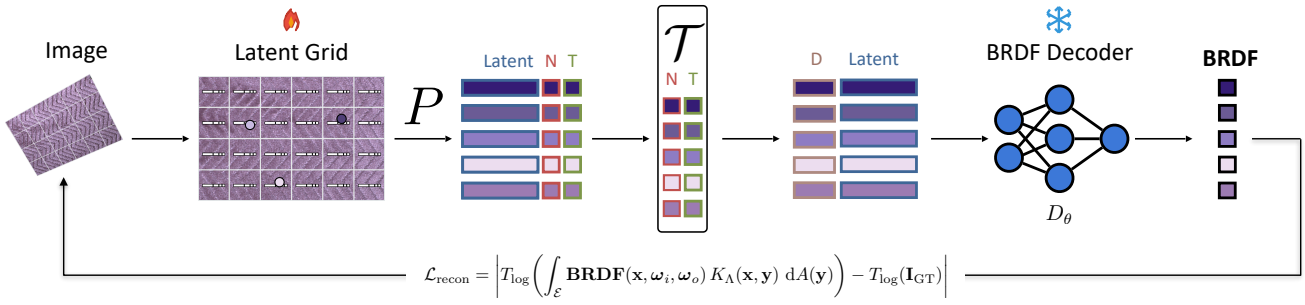

Figure 3. **Stage 2 — dense material reconstruction.** For a single target material, a dense latent-geometry texture is optimized (flame) through the parallax-aware lookup P, the shared transform $\mathcal{T}$, and the frozen decoder (snowflake); see Section 4.3. Notation as in Figure 2.

the same $\mathcal{T}$ and frozen decoder as in Stage 1. A progressive decaying Gaussian blur is applied to $\mathbf{Z}$ during optimization, acting as a coarse-to-fine regularizer that suppresses high-frequency overfitting early in training without an explicit penalty term. The detailed implementation is provided by `LatentTexture.apply_gaussian_blur` in `model/neural_brdf_refactored.py` of the supplementary code.

Texture-space displacement. As discussed in NeuMIP [13], under a flat-surface assumption, when the mesostructure has non-negligible height the point actually visible along a given direction is laterally displaced from the ray-surface intersection coordinate. Thus, similar to NeuMIP, Stage 2 includes a *parallax-aware texture lookup* P (Figure 3): a small per-material geometry MLP predicts a view- and light-dependent texture-space displacement $\Delta\mathbf{u}(\mathbf{u}, \boldsymbol{\omega}_i, \boldsymbol{\omega}_o)$ from the geometry latent at $\mathbf{u}$ and warps the coordinate before the point attributes are bilinearly sampled from $\mathbf{Z}$. The geometry MLP and $\mathbf{Z}$ are optimized jointly while D_θ stays frozen; the exact lookup is given in Section 10 and Figure 7. The same renderer and log-relative loss from Stage 1 are used throughout. For the sparse baseline datasets, which provide no shared surface parametrization, Stage 2 reverts to the per-point representation of Stage 1 (Section 10.1).

5. Experiments

To measure the value of different training corpora for Stage 1, we use a controlled experimental protocol: the decoder architecture, renderer, and Stage 2 procedure are held fixed across all comparisons, so differences in reconstruction quality are *directly* attributable to the training data. We target two key questions: *Does training on our captured dataset produce a stronger material prior than training on existing reflectance datasets?* *Does the learned prior transfer to unseen cloth materials from an independent capture system?* The complete implementation of both stages is publicly available at https://github.com/theialab/robocloth, so that every experiment reported below can be reproduced.

5.1. Datasets and Baselines

We compare three Stage 1 training sources, all consumed through the same auto-decoder loop: `RoboCloth` (our real robotic captures), `Bonn` [20] (the UBOFAB19 fabric SVBRDFs), and `MERL` [18] (real, measured isotropic BRDFs). Per-source statistics and loading details are in Section 10.1 and Table 5.

Baseline (PBR). As a non-neural reference we fit spatially-varying Disney principled BRDF [2] texture maps with the same renderer and optimizer as Stage 2. The model sums an anisotropic GGX specular lobe, a Burley diffuse lobe, and a sheen lobe; the optimized parameters are 2048×2048 textures of base color, ambient occlusion, roughness, metallic, IOR, anisotropy (strength and rotation), specular tint, sheen, and sheen tint, plus a normal map, at the same resolution as the neural latent grid. The full BRDF and its component definitions are given in Section 11.

5.2. Implementation Details

The decoder architecture, the direction encoding, the per-point and dense representations, the two-stage optimizers, and the full training and fitting schedules are reported in Section 12. The complete code for both stages is publicly released, so that all reported experiments can be reproduced and fine-grained technical questions can be answered directly from the implementation.

5.3. Results

All evaluations in this section are based on Stage-2 fitting results: the only metric available in Stage 1 is the training loss on the different training datasets, so the Stage-2 fitting is needed to evaluate the decoder on the test set. We evaluate the learned material prior in three regimes, varying only the Stage 1 training source while holding the decoder, renderer, and Stage 2 fixed: *cross-dataset transfer* to an

	Decoder training source			Baseline	
Material	RoboCloth	Bonn	MERL	PBR	vs. best
Fabric02	**34.57**	28.56	27.88	30.96	+3.61
Fabric04	**34.00**	28.50	26.66	30.09	+3.91
Fabric09	**39.50**	36.08	33.96	36.96	+2.54
Fabric11	**34.33**	29.45	27.46	31.24	+3.09
Felt01	**38.76**	33.60	32.81	35.05	+3.71
Felt03	**40.30**	34.20	33.27	36.66	+3.64
Felt05	**35.39**	32.13	30.32	32.63	+2.76
Felt10	**44.36**	38.52	36.38	41.10	+3.26
Carpet02	**38.87**	34.82	32.56	33.75	+4.05
Carpet07	**34.61**	30.17	28.85	29.48	+4.44
Carpet09	**33.17**	31.23	29.42	29.97	+1.94
Carpet12	**33.25**	29.93	28.13	28.70	+3.32
Average	**36.76**	32.27	30.64	33.05	+3.71

Table 2. **Cross-dataset transfer to UBO2014.** Decoder columns differ only in Stage-1 training source; PSNR in dB (↑). The last column reports the margin of RoboCloth over the best competing column (positive means RoboCloth leads).

	Decoder training source			Baseline	
Material	RoboCloth	Bonn	MERL	PBR	vs. best
In-domain: RoboCloth test set (held-out)					
226	**29.24**	25.63	24.24	25.35	+3.61
314	**26.60**	24.42	22.93	23.98	+2.18
370	**29.71**	29.21	27.75	29.29	+0.42
145	**28.44**	24.45	23.39	24.28	+3.99
452	**34.15**	32.91	30.89	32.06	+1.24
Average	**29.63**	27.32	25.84	26.99	+2.31
Cross-domain: Bonn test set (held-out)					
318	44.38	**45.83**	40.26	43.16	−1.45
377	16.29	**16.78**	15.57	15.93	−0.49
32	15.31	**16.42**	14.18	15.17	−1.11
226	24.14	**25.09**	22.46	23.92	−0.95
37	17.03	**18.22**	16.36	16.75	−1.19
Average	23.43	**24.47**	21.77	22.99	−1.04

Table 3. **Per-material reconstruction.** *Top:* in-domain on our held-out test set. *Bottom:* cross-dataset on held-out Bonn materials. The last column reports the margin of RoboCloth over the best competing column, in dB; positive means RoboCloth leads.

independent capture system (UBO2014), *in-domain reconstruction* on held-out materials from our own test set, and *cross-dataset comparison* on held-out Bonn materials. In every table the three decoder columns differ only in their Stage-1 training source (RoboCloth / Bonn / MERL); PBR is the analytic Disney principled fitting baseline. We report per-material PSNR↑ (dB). Fitting a dense latent grid at high resolution takes a long time (up to one day of training on an RTX 6000 Ada). Thus, due to computation-resource limits, we randomly pick a subsample of materials for evaluation: 12 from UBO2014, 5 from our own held-out test set, and 5 from the Bonn test set.

Cross-dataset transfer to UBO2014 – Table 2. We freeze each pretrained decoder and fit it to UBO2014. Because UBO2014 provides sparse per-point ray-camera observations rather than a continuously parametrized surface, we use the Stage-1-style per-point latent representation together with the per-channel scale β (Section 12), without the dense latent-geometry grid or parallax-aware texture lookup P (Section 4.3). We evaluate 12 materials randomly selected from the UBO2014 dataset [39]. UBO2014 provides $151\times151 \approx 2.3\times10^4$ angular samples per spatial pixel at 400×400 resolution. The decoder trained on our captures attains the highest PSNR on *every one* of the 12 materials, leading (by a large margin) the next-best baseline (PBR) by 3.7 dB and the strongest competing decoder (Bonn) by 4.5 dB on average. This is despite never seeing a UBO2014 sample during Stage 1, and it is direct evidence that the flexible, calibrated view-light coverage of our captures yields a more transferable cloth prior than existing measured (Bonn, MERL) or analytic (PBR) alternatives.

Qualitative comparison. Relighting the held-out UBO2014 materials in a common scene (same mesh, UV tiling, lighting, and camera for every method, so any visible difference is due *solely* to the learned material) shows the same ordering as the table: the MERL-trained decoder, trained only on spatially-uniform isotropic BRDFs, loses the weave structure and drifts in color; the Bonn-trained decoder recovers texture but shifts the overall hue, and the analytic PBR baseline misses the fine sheen; only our decoder matches the ground-truth color while preserving both the specular highlight at the folds and the fuzzy weave detail, and it attains the highest image-space PSNR on every rendered view. All qualitative results — renders of all 12 UBO2014 materials, the per-test-set reconstructions, and image-space comparisons against the captured ground truth — are provided on the project page (https://theialab.github.io/robocloth/).

In-domain SVBRDF – Table 3 (top). We next evaluate dense reconstruction on materials held out from our own dataset: for each material we freeze the decoder, optimize a dense latent grid on a subset of captured views, and evaluate on held-out views and lighting. As shown in Table 3, top, the RoboCloth-trained decoder wins on every material, confirming that the prior is well matched to our capture distribution.

Cross-domain SVBRDF – Table 3 (bottom). Finally, we test transfer to the Bonn distribution. As expected, the Bonn-trained decoder ranks first here, since these materials are held out from its own training distribution and the test set is therefore in-distribution for it. Our decoder is a consistent *second*, ahead of both the MERL-trained decoder and the PBR baseline on every material: even off-distribution, the prior learned from our captures transfers more effectively than the other non-native models. The ranking Bonn > RoboCloth > PBR > MERL holds on all five materials. Crucially, this transfer is *asymmetric*: on Bonn's own distribution our

decoder trails the native model by only 1.0 dB (23.43 vs. 24.47), whereas on our distribution the Bonn-trained decoder trails ours by 2.3 dB (27.32 vs. 29.63)—more than twice the gap. A decoder trained on RoboCloth therefore comes close to matching a Bonn-native decoder on Bonn, while *the converse does not hold*.

Ablations. We ablate the Stage-1 training-set size: retraining the decoder on random $100/300/500$-material subsets ($80/240/400$ training materials), and re-running the UBO2014 transfer shows PSNR rising consistently with more data and beginning to saturate at our full corpus. Please refer to the supplementary Section 8, Fig. 5, and Tab. 4.

6. Conclusion

We presented RoboCloth, a dataset of 500 real cloth materials captured under flexible, calibrated camera and light coverage using a robotic acquisition system. Our end-to-end pipeline produces per-material training records — aligned HDR images, sparse surface geometry, and structured observation tensors that feed directly into neural decoder training without manual annotation. Under a controlled two-stage protocol in which only the Stage 1 training data varies, a decoder trained on our captures learns a stronger cloth prior than decoders trained on existing measured reflectance datasets, recovering more faithful held-out appearance and transferring competitively to the UBO2014 benchmark.

Limitations. The dataset is cloth-specific: the acquisition protocol and evaluation are tailored to near-planar fabric samples and do not generalize to three-dimensional objects or non-textile materials. Acquisition throughput is modest: each material takes $\approx$20 minutes and needs a human operator to mount and change the sample between sessions. Thus, although in practice we have $\approx$5000 more cloth samples, throughput is currently limited by the most difficult resource to scale: the graduate student. We are therefore considering a different capture modality for faster acquisition; throughput could also clearly be scaled by adding more robot arms. Notably, the dataset-size ablation (Figure 5 in Section 8) shows that transfer PSNR still improves at the full corpus: learning has not saturated, and we stopped at 500 materials simply as a round number, given the slow capture. Since a material prior is available from previous scans, an active-capture agent could select the next-best view-light configuration for each new sample, replacing the $\approx$580 uniformly scheduled frames with a small adaptive set and directly addressing the throughput bottleneck. Finally, this work performs SVBRDF inversion for planar materials. How to integrate the resulting material priors effectively within a *scene-level* 3D Gaussian Splatting reconstruction [11] remains an open research direction.

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RoboCloth: A Large-Scale Real Cloth Material Dataset for Neural Reflectance Reconstruction

Supplementary Material

Supplementary Overview

This supplementary material is organized as follows. Section 7 presents the grazing-angle supervision ablation and Section 8 the dataset-size ablation. Section 9 provides full details of the dataset construction pipeline, including the capture trajectory design, complete details of the hand-eye and turntable-axis calibration, the Umeyama similarity alignment, and the radiometric calibration procedures summarized in the main paper. Section 10 expands the two pipeline operators—the coordinate transform $\mathcal{T}$ and the parallax-aware texture lookup P—that appear as labeled boxes in the main-paper pipeline figures (Figs. 2 and 3). Section 11 gives the full Disney principled BRDF used by the PBR fitting baseline. Section 12 reports the full set of implementation hyperparameters for Stage 1 and Stage 2, together with additional notes on the 8-bit Adam optimizer, the per-point optimizable parameters, and the round-robin material sampling strategy. Section 13 documents the schema of the released Hugging Face dataset. Additional qualitative results on held-out materials and UBO2014 transfers are provided on the project page (https://theialab.github.io/robocloth/).

7. Ablation: Grazing-Angle Supervision

Our captured observations do not cover near-grazing incidence ($\cos\theta_i \approx 0$), leaving the decoder unconstrained there. The grazing-decay loss of (3) constrains the *shape* of the contribution $f_r \cos\theta_i$ to fall off smoothly to zero as $\theta_i \to 90°$, without forcing the raw BRDF f_r itself to vanish. We compare it against two simpler pseudo-observation losses and a no-grazing baseline. **ZeroExact** adds a pseudo-observation that pins the raw BRDF to zero at *exactly* the grazing angle ($\theta_i = 90°$). **NearZeroBRDF** pins the raw BRDF to zero over a thin band just below grazing ($\cos\theta_i \in [0.005, 0.03]$). **No grazing** applies no such supervision.

To visualize the learned reflectance we use the Stage-1 BRDF-lobe plot: we select one learned point latent, freeze the decoder, fix the outgoing direction $\boldsymbol{\omega}_o$ (one colored curve per $\theta_o \in \{15, 30, 45, 60\}°$), and sweep the incident elevation θ_i from 0° (surface normal) to ±90° (grazing) at a fixed azimuth; the polar radius is the contribution $f_r \cos\theta_i$ and the dashed line marks the specular direction. Figure 4 shows four representative surface points.

Only our loss ((3)) yields lobes that taper smoothly to zero at grazing. **ZeroExact** fixes only the single, measure-zero point $\theta_i = 90°$ and leaves its entire neighborhood free, so the contribution balloons outward toward grazing with no smooth transition; forcing f_r itself to zero is also unphysical, since the raw BRDF is generally non-zero near grazing — it is the *product* $f_r \cos\theta_i$ that must vanish. **NearZeroBRDF** pins f_r to zero across a thin near-grazing band but cannot shape the transition between the last supervised observation and the grazing region, leaving non-physical lobes and spikes near ±90°. The no-grazing baseline is unconstrained and shows the same inflation.

8. Ablation: Dataset Size

We ablate the size of the Stage-1 training corpus to test whether our dataset provides a sufficiently diverse distribution that the learned prior keeps improving as more materials are added. We draw random subsets of 100, 300, and 500 cloth materials and apply the same $0.8/0.2$ split to each, yielding 80, 240, and 400 training materials respectively (denoted RoboCloth-100/300/500). Every subset trains a decoder for the *same* number of Stage-1 iterations (4.8×10^5 steps) under the identical architecture and loss, and is then evaluated with the Stage-2 UBO2014 transfer protocol of Section 5.3: we freeze the decoder and optimize only the per-point latent and the per-channel scale $\boldsymbol{\beta}$ on the same 12 UBO2014 materials for 60 epochs.

Figure 5 and Table 4 report the mean UBO2014 PSNR over all 12 transfer materials. Transfer PSNR increases monotonically with the training-set size ($36.28 \to 36.57 \to 36.76$ dB), and the per-material numbers improve almost everywhere: the decoder consistently benefits from more of our data, indicating that the corpus carries genuine appearance diversity rather than redundant samples. The gain also flattens toward the full set (+0.29 dB from 80 to 240 materials versus +0.19 dB from 240 to 400), suggesting that our full corpus is approaching saturation for this transfer task.

9. Dataset Construction: Full Details

9.1. Capture Trajectory

The camera viewpoints are drawn from a hemisphere above the sample, with polar angle $\phi \in [0, 2\pi/5]$ and two azimuthal lobes at $\theta = 0$ and $\theta = \pi$. The hemisphere is partitioned into 8 sectors visited in a space-filling order to minimize arm travel. Light positions are sampled by ray-bounding-box intersection on a wireframe cage above the sample: $N_{\text{base}} = 50$ base positions cover the upper hemisphere uniformly, and $N_{\text{tail}} = 8$ additional positions provide

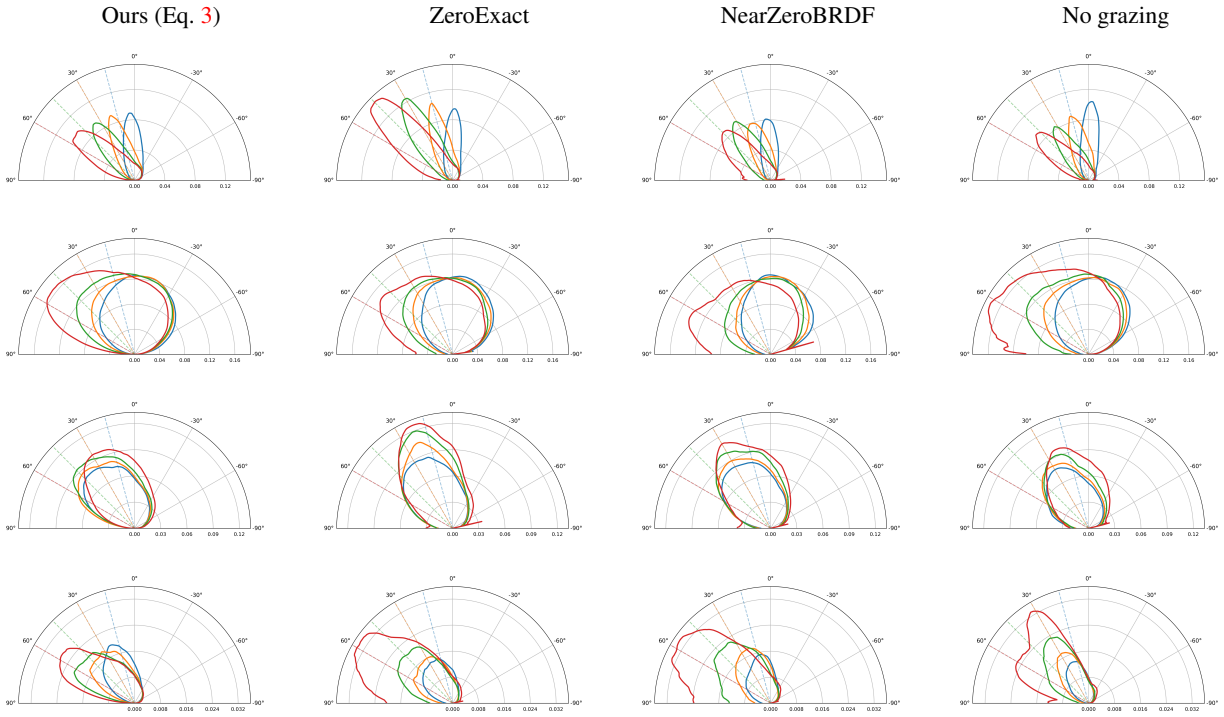

Figure 4. **Grazing-angle supervision.** Polar BRDF-lobe plots of the learned contribution $f_r \cos\theta_i$ vs. incident elevation θ_i ($0°$=normal at top, $\pm 90°$=grazing at the sides); each colored curve is a fixed outgoing direction (blue $\theta_o = 15°$, orange $30°$, green $45°$, red $60°$), and the dashed line marks the specular direction. Each row is a different surface point; columns are the four grazing-supervision variants. Only our loss (first column, Eq. 3) makes $f_r \cos\theta_i$ fall off smoothly to zero near grazing; ZeroExact, NearZeroBRDF and the no-grazing baseline all leave the near-grazing region uncontrolled, inflating the lobes toward $\pm 90°$.

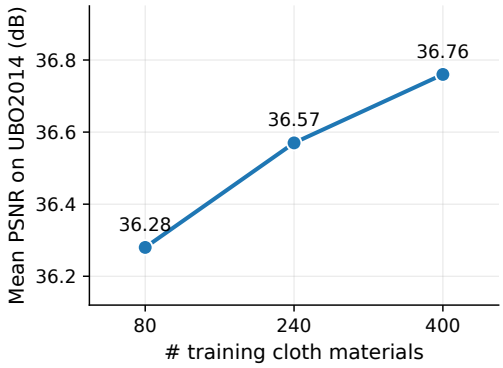

Figure 5. **Transfer PSNR vs. training-set size.** Mean UBO2014 transfer PSNR (over all 12 transfer materials) against the number of training cloth materials (80/240/400, i.e. the 100/300/500-material subsets). Accuracy rises consistently with more data and begins to saturate at our full corpus. The vertical axis is zoomed to make the trend visible.

Material	RoboCloth-100 (80)	RoboCloth-300 (240)	RoboCloth-500 (400)
fabric02	34.38	**34.60**	34.57
fabric04	33.33	33.72	**34.00**
fabric09	38.93	39.40	**39.50**
fabric11	33.76	34.04	**34.33**
felt01	38.38	38.70	**38.76**
felt03	39.75	40.29	**40.30**
felt05	34.82	35.05	**35.39**
felt10	43.62	43.91	**44.36**
carpet02	38.22	38.58	**38.87**
carpet07	34.22	34.37	**34.61**
carpet09	32.97	33.05	**33.17**
carpet12	33.01	33.11	**33.25**
Average	36.28	36.57	**36.76**

Table 4. **Dataset-size ablation (UBO2014 transfer).** Per-material PSNR (dB, ↑) for decoders trained on the 100/300/500-material subsets (80/240/400 training materials after the 0.8/0.2 split). PSNR rises consistently with more training data.

finer angular resolution near grazing. Between consecutive frames the turntable advances by 29 ± 19 degrees, with a uniform $[0, 360)$ initial rotation drawn once per material.

Reproducibility is ensured by a `NumPy SeedSequence` keyed on the cloth identifier: light positions are shared across

all materials in a 10-batch, while turntable jitter and per-step camera ϕ jitter are unique per material.

9.2. Hand-Eye and Turntable-Axis Calibration

Hand-eye calibration. The hand-eye transform ${}^{g}T_{c}$ is recovered from a sequence of ChArUco-board [6, 29] observations using Tsai's formulation [32], as implemented in `cv2.calibrateHandEye`. For every frame the board pose ${}^{c}T_{t,i}$ is estimated with `solvePnP` from the detected board corners; pairing these target-to-camera poses with the robot gripper-to-base poses $\{{}^{b}T_{g,i}\}$ gives the $AX = XB$ system that we solve for ${}^{g}T_{c}$. The calibration is performed once per camera mounting and stored as a fixed transform for all subsequent captures.

Turntable-axis calibration. The turntable rotation axis $(\hat{\mathbf{n}}, \mathbf{c})$ is calibrated from a capture scan itself, by aligning the COLMAP camera centers $\{\mathbf{p}_i\}$ with the robot-derived camera centers $\{\mathbf{q}_i\}$ obtained from the logged gripper poses and the hand-eye transform. Writing $R_i = R(\hat{\mathbf{n}}, \theta_i)$ for the rotation by the logged turntable angle θ_i, we alternate three steps to convergence: (i) a Umeyama similarity fit [34] of $\{\mathbf{q}_i\}$ onto the centers predicted by the current $(\hat{\mathbf{n}}, \mathbf{c})$, giving $(s, R, \mathbf{t})$; (ii) a linear least-squares update of the axis point from the stacked system $(I - R_i)\,\mathbf{c} = sR\,\mathbf{q}_i + \mathbf{t} - R_i\,\mathbf{p}_i$; and (iii) a Levenberg–Marquardt step on $\hat{\mathbf{n}}$, parameterized in its tangent space so that it stays a unit vector. A final polish freezes the similarity and refines $(\hat{\mathbf{n}}, \mathbf{c})$ alone under a Huber loss. Turntable angles are counter-clockwise about $\hat{\mathbf{n}}$ by the right-hand rule, which fixes the $\hat{\mathbf{n}} \equiv -\hat{\mathbf{n}}$ sign convention.

Derotation derivation. For completeness we derive the derotation operator used in Section 3.2. Let $R(\hat{\mathbf{n}}, \alpha)$ denote the Rodrigues rotation by angle α about $\hat{\mathbf{n}}$, and define

$$\mathcal{T}(R, \mathbf{c}) = \begin{bmatrix} R & (I - R)\mathbf{c} \\ \mathbf{0} & 1 \end{bmatrix} \in \mathrm{SE}(3),$$

which rotates by R about the point $\mathbf{c}$. The derotation transform that undoes turntable angle θ_i is

$${}^{w_0}T_{b,i} = \mathcal{T}\big(R(\hat{\mathbf{n}}, -\theta_i),\, \mathbf{c}\big),$$

and the camera pose in the common 0° world frame is

$${}^{w_0}T_{c,i} = {}^{w_0}T_{b,i}\,{}^{b}T_{g,i}\,{}^{g}T_{c}.$$

9.3. Similarity Alignment: Umeyama Derivation

Let $\{\mathbf{p}_i\}$ be the COLMAP camera centers and $\{\mathbf{q}_i\}$ the corresponding robot-derived centers in the 0° frame. We solve for $(s, R, \mathbf{t}) \in \mathrm{Sim}(3)$ such that $\mathbf{q}_i \approx sR\mathbf{p}_i + \mathbf{t}$. Define

$$\bar{\mathbf{p}} = \tfrac{1}{N}\textstyle\sum_i \mathbf{p}_i, \quad \bar{\mathbf{q}} = \tfrac{1}{N}\textstyle\sum_i \mathbf{q}_i, \quad \Sigma = \tfrac{1}{N}\textstyle\sum_i (\mathbf{q}_i - \bar{\mathbf{q}})(\mathbf{p}_i - \bar{\mathbf{p}})^\top.$$

The SVD $\Sigma = UDV^\top$ yields

$$R = USV^\top, \quad S = \mathrm{diag}(1, 1, \det(UV^\top)),$$
$$s = \frac{\mathrm{tr}(DS)}{\frac{1}{N}\sum_i \|\mathbf{p}_i - \bar{\mathbf{p}}\|^2}, \quad \mathbf{t} = \bar{\mathbf{q}} - sR\bar{\mathbf{p}}.$$

The correction matrix S handles the reflection ambiguity in the SVD. The mean alignment residual $\frac{1}{N}\sum_i \|sR\mathbf{p}_i + \mathbf{t} - \mathbf{q}_i\|$ is logged per material as a reconstruction quality diagnostic; frames with per-image residual exceeding $\tau = 16\,\mathrm{mm}$ are discarded before the observation tensor is assembled.

9.4. Radiometric Calibration Details

Demosaicing and quantum-efficiency correction. The Spinnaker sensor outputs a 14-bit Bayer mosaic in RGGB pattern, stored left-aligned in a 16-bit PNG. We demosaic using the Menon 2007 algorithm [19] and apply a channel-wise quantum-efficiency correction $(g_R, g_G, g_B) = (1.5806, 1.0, 1.0659)$ before clipping to the 16-bit range. Both the raw mosaic and the debayered HDR PNG are retained on disk.

Camera factor calibration. The camera factor κ converts pixel values to physical radiance under the linear sensor model. It is measured once per exposure cohort by imaging a near-Lambertian white reference card while sweeping the LED through a fixed polar grid (`linear_hemi` routine). The predicted irradiance at each step is known analytically from the robot pose and the emitter model $\Lambda(\theta)$; fitting the slope of measured intensity against predicted radiance yields κ. For our `Dataset_Nov11` captures, three exposure cohorts $\{20000, 20000, 8000\}\,\mu$s give factors $(\kappa_1, \kappa_2, \kappa_3)$ satisfying $\kappa_3 = \kappa_2 \cdot (8000/20000)$ to within the fit residual.

Emitter directional distribution. For every observation of the white reference card we compute the ratio $r = I^{\mathrm{iso}}_{\mathrm{pred}}/I_{\mathrm{meas}}$, bin by the LED-emission angle θ in 1° bins over $[0^\circ, 90^\circ]$, average within each bin, fill empty bins by nearest-neighbor interpolation, and normalize by the maximum. The result is a single 90-entry table $\Lambda(\theta) \in [0, 1]$, shared across the three colour channels.

Color correction matrix. A 3×3 CCM is estimated from a 24-patch ColorChecker. Reference values are converted from manufacturer $\mathrm{Lab}(D_{50})$ to linear $\mathrm{sRGB}(D_{65})$, where D_{50} and D_{65} are the two spaces' reference white illuminants, via

$$\mathrm{Lab}(D_{50}) \to \mathrm{XYZ}(D_{50}) \xrightarrow{\text{Bradford CAT}} \mathrm{XYZ}(D_{65}) \to \text{linear } \mathrm{sRGB}(D_{65}),$$

and the matrix $M = \arg\min_M \|CM - R\|_F^2$ is solved in closed form, where $C \in \mathbb{R}^{24\times 3}$ holds captured patch colors and $R \in \mathbb{R}^{24\times 3}$ the references. Color accuracy is assessed with the CIEDE2000 color difference ΔE_{00} [16] on the

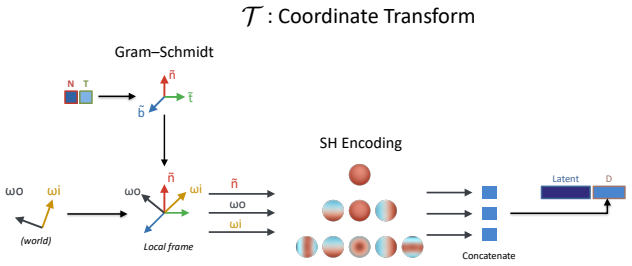

Figure 6. **The coordinate-transform operator $\mathcal{T}$.** A per-point (Stage 1) or per-texel (Stage 2) normal $\mathbf{n}$ and tangent $\mathbf{t}$ are orthogonalized by Gram–Schmidt into a TBN frame $(\tilde{\mathbf{n}}, \tilde{\mathbf{t}}, \tilde{\mathbf{b}})$. The world-space incident and outgoing directions $\boldsymbol{\omega}_i, \boldsymbol{\omega}_o$ are rotated into this local frame; the local directions together with $\tilde{\mathbf{n}}$ are then encoded with degree-3 real spherical harmonics and concatenated with the latent code before being passed to the BRDF decoder.

same 24 patches. The CCM is shipped alongside the data but not applied to the released images; see Section 3.3 for the rationale.

10. Pipeline Module Details

The two operators that appear as labeled boxes in the main-paper pipeline (Figs. 2 and 3) are expanded here: the coordinate transform $\mathcal{T}$ (Figure 6), shared by both stages, and the parallax-aware texture lookup P (Figure 7), used in Stage 2.

Construction of $\mathcal{T}$. Given a point's normal $\mathbf{n}$ and tangent $\mathbf{t}$, $\mathcal{T}$ orthonormalizes a TBN frame by Gram–Schmidt,

$$\tilde{\mathbf{n}} = \hat{\mathbf{n}}, \quad \tilde{\mathbf{t}} = \mathbf{t} - \widehat{(\mathbf{t} \cdot \tilde{\mathbf{n}})}\,\tilde{\mathbf{n}}, \quad \tilde{\mathbf{b}} = \tilde{\mathbf{n}} \times \tilde{\mathbf{t}},$$

rotates the incident and outgoing directions into this frame with $R = [\tilde{\mathbf{t}}\ \tilde{\mathbf{b}}\ \tilde{\mathbf{n}}]$, and encodes the rotated directions together with $\tilde{\mathbf{n}}$ by degree-3 real spherical harmonics (48 dims), concatenating the latent code $\mathbf{z}$ to form the decoder input. The orthogonalization is differentiable, so gradients flow back into the unconstrained $(\mathbf{n}, \mathbf{t})$ tables without a projected-gradient step.

Construction of P. In Stage 2, let $[\cdot]_{\text{geom}}$ select the geometry channels of the latent-geometry grid. The initial lookup and learned displacement are

$$\mathbf{g}(\mathbf{u}) = \big[\text{BilinearSample}(\mathbf{Z}, \mathbf{u})\big]_{\text{geom}},$$
$$\Delta\mathbf{u} = g_\phi(\mathbf{g}(\mathbf{u}), \boldsymbol{\omega}_i, \boldsymbol{\omega}_o) \in \mathbb{R}^2.$$

The parallax-aware texture lookup then returns the complete point attributes from the displaced coordinate,

$$P(\mathbf{Z}, \mathbf{u}, \boldsymbol{\omega}_i, \boldsymbol{\omega}_o) = \text{BilinearSample}\big(\mathbf{Z},\ \mathbf{u} + \Delta\mathbf{u}\big).$$

The displacement shifts the sampled texel with view and light so the per-texel attributes follow the visible yarn surface; setting $\Delta\mathbf{u} = \mathbf{0}$ recovers a standard bilinear lookup.

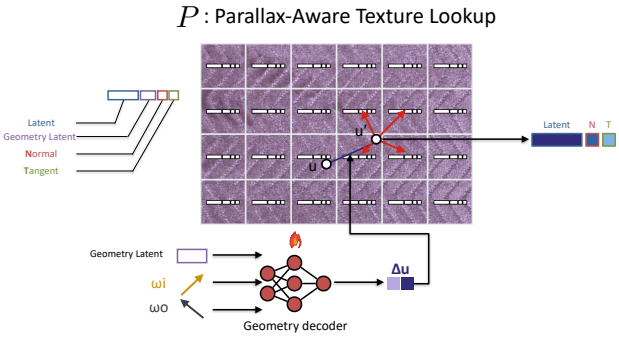

Figure 7. **Parallax-aware texture lookup P (Stage 2).** An initial sample at $\mathbf{u}$ provides the geometry latent $\mathbf{g}(\mathbf{u})$. Together with the incident and outgoing directions $\boldsymbol{\omega}_i, \boldsymbol{\omega}_o$, a small geometry decoder maps it to a texture-space displacement $\Delta\mathbf{u}$. The final point attributes are bilinearly sampled from the latent-geometry grid $\mathbf{Z}$ at $\mathbf{u}' = \mathbf{u} + \Delta\mathbf{u}$, reproducing yarn-level parallax without an explicit height field.

Source	Data type	# Materials
RoboCloth	Real robotic captures (HDR)	∼400
Bonn	Multi-LED and linear-light-source	311
MERL	Measured isotropic BRDFs	100

Table 5. Training sources for the neural decoder (Stage 1). Each RoboCloth material contributes ∼580 HDR frames. Every Bonn material is measured in three modalities: PAN (panchromatic, 24 unfiltered LEDs, 388 images), POLY (polychromatic, RGB-filtered LEDs, 100 images), and LLS (linear light source, 14 angles, 280 images). MERL supplies tabulated BRDFs rather than images and is sampled directly.

10.1. Stage-1 Training Sources

RoboCloth is loaded from the structured $(K, V, 3)$ `uint16` tensors described in Section 3.4. All retained materials are held in CPU memory and sampled uniformly across (material, point, view) triples. **Bonn** [20] is loaded from the released AxF files. Here the "Bonn" source is UBO-FAB19 [20], the fabric release (v1) of the Bonn SVBRDF dataset; we use "Bonn" as shorthand for it throughout. PAN and POLY observations are fed to the renderer with a single emitter sample ($S = 1$) using the published per-LED point-light positions. LLS observations are integrated with $S = 4$ Monte Carlo samples drawn from the four corners of each linear light source. **MERL** [18] is a real, measured source: we render each of the 100 isotropic BRDFs from the Rusinkiewicz-parameterized table at random $(\theta_h, \theta_d, \phi_d)$ triples. The renderer degenerates to a direct BRDF lookup — no emitter, no inverse-square fall-off — and the loss reduces to a direct fit on tabulated values.

When these sparse sources (and the UBO2014 test set) are used as Stage 2 reconstruction targets, they are fit with

the same per-point latent representation as Stage 1—one $(\mathbf{z}, \mathbf{n}, \mathbf{t})$ triplet per observation point, optimized with the frozen decoder and the per-channel scale $\boldsymbol{\beta}$—rather than the dense latent-geometry grid and parallax-aware texture lookup P, which require the continuously parametrized surface available only for our own captures.

11. Disney Principled PBR Baseline

The non-neural reference of Section 5.1 fits, per material, a set of Disney principled BRDF [2] parameter maps with the same renderer and optimizer as Stage 2. At each shading point the BRDF is the sum of an anisotropic microfacet specular lobe, a Burley diffuse lobe, and a sheen lobe,

$$f_r^{\text{PBR}} = f_{\text{spec}} + f_{\text{diff}} + f_{\text{sheen}}, \tag{4}$$

evaluated in the local TBN frame with half-vector $\mathbf{h} = \widehat{\boldsymbol{\omega}_i + \boldsymbol{\omega}_o}$. We write $\text{lerp}(a, b, t) = (1-t)\,a + t\,b$.

Specular. The specular lobe is the standard anisotropic microfacet term

$$f_{\text{spec}} = \frac{D(\mathbf{h})\,G(\boldsymbol{\omega}_i, \boldsymbol{\omega}_o)\,F(\boldsymbol{\omega}_o \cdot \mathbf{h})}{4\,(\mathbf{n}\cdot\boldsymbol{\omega}_i)\,(\mathbf{n}\cdot\boldsymbol{\omega}_o)}, \tag{5}$$

where D and G are the anisotropic GGX normal-distribution and Smith masking-shadowing terms, evaluated with directional roughnesses $\alpha_x = \rho\,(1+a)$ and $\alpha_y = \rho\,(1-\frac{1}{2}a)$ for roughness ρ and anisotropy strength a, with the tangent frame rotated in-plane by a learned anisotropy angle. F is the Schlick Fresnel term $F = C_{\text{spec}} + (1 - C_{\text{spec}})\,(1 - \boldsymbol{\omega}_o \cdot \mathbf{h})^5$ with a tinted dielectric reflectance

$$C_{\text{spec}} = \text{lerp}\Big(F_0^{\text{d}}\ \text{lerp}(\mathbf{1}, C_{\text{tint}}, s_{\text{spec}}),\ \mathbf{c},\ \mu\Big), \tag{6}$$

where $\mathbf{c}$ is the base color, μ the metallic factor, s_{spec} the specular tint, $F_0^{\text{d}} = \left(\frac{\eta-1}{\eta+1}\right)^2$ the dielectric reflectance for IOR η, and $C_{\text{tint}} = \mathbf{c}/(0.3\,c_R + 0.6\,c_G + 0.1\,c_B)$ the luminance-normalized hue.

Diffuse. The diffuse lobe is Burley's roughness-dependent model,

$$f_{\text{diff}} = \frac{\mathbf{c}\,(1-\mu)\,\text{ao}}{\pi}\,F_d(\boldsymbol{\omega}_i)\,F_d(\boldsymbol{\omega}_o), \tag{7}$$

with $F_d(\boldsymbol{\omega}) = 1 + (F_{90} - 1)\,(1 - \mathbf{n}\cdot\boldsymbol{\omega})^5$, $F_{90} = 0.5 + 2\rho\,(\boldsymbol{\omega}_i \cdot \mathbf{h})^2$, and ambient occlusion ao.

Sheen. The sheen lobe adds the characteristic fabric edge glow,

$$f_{\text{sheen}} = (1 - \boldsymbol{\omega}_i \mathbf{h})^5\,w_{\text{sheen}}\,(1-\mu)\ \text{lerp}(\mathbf{1}, C_{\text{tint}}, s_{\text{sheen}}), \tag{8}$$

with sheen strength w_{sheen} and sheen tint s_{sheen}. The optimized parameters are per-texel maps of base color $\mathbf{c}$, ambient occlusion, roughness, metallic, IOR, anisotropy (strength and rotation), specular tint, sheen, and sheen tint, together with a normal map, all at 2048×2048 to match the neural latent grid.

12. Implementation Details

Full hyperparameters for Stage 1 and Stage 2 are reported in Table 6. This section gives the full implementation.

Architecture and optimization. The decoder is a four-layer MLP (width 256, `LeakyReLU`(0.2), skip connection at layer 2, softplus output); incident and outgoing directions use real spherical harmonics of degree $L = 3$ and the latent dimension is $d = 24$. Stage 1 jointly trains the shared decoder and the per-point parameter tables for all ≈ 400 materials with 8-bit Adam; Stage 2 freezes the decoder and fits each held-out material with full-precision Adam. Owing to compute and memory limits, every material is trained and fit on a fixed random 10% subsample of its surface points, applied identically to all sources so the per-material budget is equal. In every Stage-2 fit we additionally optimize a per-channel scale $\boldsymbol{\beta}$ that aligns the frozen decoder to the target capture's color space and radiometric calibration.

Per-point optimizable parameters. In Stage 1, each sparse surface point owns three directly optimizable quantities stored as `requires_grad=True` tensors: the latent code $\mathbf{z}_{m,j}$, the surface normal $\mathbf{n}_{m,j}$, and the tangent $\mathbf{t}_{m,j}$. These are updated by the same Adam step as θ, with the learning rate for $(\mathbf{n}, \mathbf{t})$ matching that of the latent tables (2×10^{-3}). The Gram–Schmidt orthogonalization is applied in the forward pass and is fully differentiable, so gradients flow back into the unconstrained $(\mathbf{n}, \mathbf{t})$ parameters without any projected-gradient step.

8-bit Adam in Stage 1. Storing the complete per-point latent tables for $\sim$400 materials with $V_m \in [10^3, 5 \times 10^3]$ points each requires keeping $\sim 400 \times 5000 \times 24 \times 4$ bytes $\approx$ 192 GB of optimizer state in full precision. We use `bitsandbytes` 8-bit Adam to reduce this by $4\times$, fitting all optimizer states on a single node without offloading.

Round-robin material sampling. At each Stage 1 iteration, the active material set is rotated by a deterministic round-robin so that every material is touched within a fixed window of iterations. Within the active set, (point, view) pairs are sampled without replacement until the batch of 5×10^5 rays is filled.

Validation and checkpointing. At validation time, emitter samples are accumulated in chunks of $S_{\text{chunk}} = 2$ to bound peak memory. We checkpoint every 10^4 steps and select the checkpoint at which the Bonn validation loss plateaus ($\approx 4.8 \times 10^5$ steps) as the Stage 2 initialization for all cross-dataset comparisons.

13. Dataset Schema

The dataset is released as a public Hugging Face dataset under the CC BY 4.0 license. Table 7 documents its repository layout: one folder per material under `materials/`, shared

Component	Setting
Direction encoding	Real spherical harmonics, $L = 3$, 48 dims total
Latent dimension	$d = 24$
Decoder MLP	$[256, 256, 256, 256]$, LeakyReLU(0.2), softplus output, skip @ layer 2
Stage 1 optimizer	8-bit Adam, η_{lat}=2×10^{-3}, η_{dec}=2×10^{-4}
Stage 1 batch	5×10^5 rays, round-robin material sampling
Stage 1 iterations	$\approx 4.8 \times 10^5$ steps
Stage 1 loss	Log-relative; ρ_{ref} set to each source's median BRDF magnitude: 0.02 (Bonn), 0.057 (MERL), 1.5×10^4 (RoboCloth)
Stage 1 Monte Carlo samples	S=4 (RoboCloth area emitter, Bonn LLS), S=1 (point lights)
Stage 2 representation	Dense latent–geometry grid $2048^2 \times 46$ (24 latent + 16 geometry + 6 TBN-frame channels), init $\mathcal{N}(0, 0.1^2)$
Stage 2 geometry	Per-pixel TBN (Gram–Schmidt) + displacement MLP $[16, 16, 8]$
Stage 2 regularization	Progressive Gaussian blur, $\sigma(t) = 2 \cdot 2^{-t/15000}$
Stage 2 optimizer	Adam (full precision), lr=2×10^{-3} (texture and geometry)
Stage 2 epochs	60 (UBO2014 transfer) / 120 (Bonn) / 80–120 (our captures, complexity-dependent)
Stage 2 loss	Log-relative, ρ_{ref}=0.05 (UBO2014 transfer)
Per-channel scale	$\boldsymbol{\beta}$, init $\mathbf{1}$
PBR baseline	Disney principled, $\mathbf{c}_b$+ρ+m+IOR+aniso+sheen+normal at 2048^2
Test SPP	16
Metrics	PSNR, log-relative loss, qualitative renderings

Table 6. Full implementation details. Settings are shared across datasets unless noted.

Path	Format	Contents
`materials/{id}/` — one folder per material		
`hdr.tar`	TAR of 16-bit PNG	∼580 cropped, polygon-masked linear-HDR frames
`observations_structured.npz`	NumPy NPZ	`rgbs` $(K, V, 3)$ `uint16` radiance tensor; `xyz` $(V, 3)$ `float32` point positions; `point_ids` $(V,)$ `int32`; `cam_pos`, `light_pos` $(K, 3)$
`point_positions.npz`	NumPy NPZ	`point_ids` $(V,)$ and `positions` $(V, 3)$ in base frame (m)
`rotated_camera.json`	JSON	Per-frame camera-to-world pose: position and 3×3 rotation (turntable-aligned)
`scan_log.json`	JSON	Per-scan acquisition log (camera and emitter ids, exposure)
`point_metadata.json`	JSON	Surface-point count V and per-material bookkeeping
`hdr_crop_bboxes.json`	JSON	Per-frame crop box used to cut each frame to the cloth
`bbox.json`	JSON	Axis-aligned bounding box of the sample (base frame, m)
`unmatched_scan_ids.json`	JSON	Scan ids excluded from cross-view matching
`globals/` — shared across all materials		
`emitter_calibration.json`	JSON	Directional emitter profile $\Lambda(\theta)$
`camera_factor.json`	JSON	Radiometric scale κ per exposure cohort
`sample_size.json`	JSON	Physical sample size (w_m, l_m) per material id
`training_list_{100,300,442,500}.txt`	text	Training-split material ids at four corpus sizes
`test_list_{100,300,442,500}.txt`	text	Held-out test material ids at four corpus sizes
Repository root		
`README.md`, `LICENSE`, `croissant.jsonld`	—	Dataset card, CC BY 4.0 license, and Croissant (RAI) metadata

Table 7. Released Hugging Face dataset schema. `{id}` is the integer material id; K and V are the per-material observation and surface-point counts (Section 3.4).

calibration and split files under `globals/`, and the dataset card, license, and Croissant metadata at the repository root. Floating-point quantities are in SI units (meters, radians) unless noted; K and V denote the per-material observation and surface-point counts introduced in Section 3.4.